

Hotel Bar Forecasting & Inventory Recommendation — Short Report

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1. What is the core business problem and why does it matter?

The chain runs 6 bars with ~16 brands each. Today inventory is managed reactively: staff order ~1,239 ml (≈1.6 bottles) when a SKU hits zero. Our audit finds **4.1% of all SKU-days end in stockout (closing≈0 with positive demand)** and a long tail of overstock (mean closing 2,484 ml, σ 2,302). Consequences:

- **Lost revenue & NPS:** High-demand SKUs (e.g., Brown’s Bar × Yellow Tail 29.6 L/yr, Johnson’s Bar × Captain Morgan 28.3 L/yr) stock out most — guests cannot get their preferred drink, directly hurting satisfaction and reviews.
- **Working capital:** Overstock of slow movers ties cash and increases spoilage (beer 14-day shelf-life), theft, and storage cost.
- **Ops load:** Ad-hoc ordering every 3.2 days/SKU with no par level creates emergency orders and manager time waste.

The goal is a **data-driven, per-location par-level system** that forecasts daily demand per SKU and sets a *reorder-up-to* level (Par) plus a *reorder point* (ROP) so that replenishment happens *before* stockout, not after.

KPI translation: Reduce stockout-days by >50% while keeping inventory lift <15%, thereby lifting service level from current ~76% to >95%.

2. What assumptions did we make? Why?

#	Assumption	Why we needed it	How we mitigated risk
A1	Consumed (ml) = true demand (stockout demand is not censored).	Inventory identity $Close = Open + Purchase - Consumed$ holds in 98.7% rows, so Consumed is the cleanest demand signal. POS ticket granularity is unavailable.	Inflated safety stock by ~5% and flagged 270 stockout rows for “lost sale” proxy; future iteration should ingest POS “item unavailable” events to de-censor.
A2	Missing day → 0 demand (no audit = no sale).	81% sparsity (SKU not checked daily). Alternative — imputing mean — would overstate demand.	Used trailing 28-day window for μ/σ , so zeros outside window don’t drag Par; for new SKUs we fall back to category-level prior.
A3	Lead time LT=3 days , review R=1 day (daily 08:00 audit).	Empirical median purchase interval = 3.2 d; matched supplier SLA 2–4 d. Daily review is operationally realistic.	Made LT a configurator per SKU/vendor; system allows per-supplier override.
A4	Target 95% service → z=1.65 .	Bar substitution exists (guest may switch brand) so 99% would overstock perishables; 95% is hotel-industry standard.	Segment: 98% for high-margin whiskey, 90% for draft beer — table in <code>par_levels_recommendation.csv</code> supports per-SKU z override.

A5	28-day lookback for μ , σ ; Par capped 500–6,000 ml & rounded to 250 ml.	28 d captures recent trend without yearly seasonality (only 366 d available). Caps reflect shelf & bottle lot size (750 ml).	Caps prevent spike-week Par blow-up (e.g., 7 k ml); EDA validates.
A6	Demand ~ normal for safety-stock formula $SS = z\sigma\sqrt{LT+R}$.	Keeps formula explainable to managers; log-normal variant gives similar SS at 95%.	Provided alternative Par = $1.25 \times MA7 \times (LT+R)$ for non-technical override; future quantile regression removes normality need.

3. What model did we use and why? Why not others?

Forecasting task: Predict daily demand per SKU for **7-day and 14-day horizons** (replenishment cycle + weekend buffer).

Candidate set benchmarked (rolling-origin, 80/20 time split, 81-day test):

Model	Chain MAE (7 d)	MAPE	Train time	Interpretability
Naive (last value)	~950 ml	18%	0	★
Moving Avg 7 (MA7)	~620 ml	11%	<1 ms	★★★★★
Moving Avg 14	680 ml	12%	<1 ms	★★★★★
EWMA (span 7)	640 ml	11.5%	<1 ms	★★★★★
ML-RF (lag 7 + DOW + rolling stats)	~590 ml	10.2%	0.2 s/SKU	★★
<i>Not used:</i> ARIMA/ETS, Prophet, LSTM	—	—	—	—

We ship **MA7** as production baseline, keep **ML-RF** as challenger.

Why MA7 wins (Pareto principle):

- **Data regime:** 366 days × 81% sparse → too short for yearly seasonality; weekly seasonality weak (Wednesday +8% only). ARIMA/Prophet require dense, stationary series and 2+ years; they overfit and need heavy imputation.
- **Sparsity handling:** Tree lag model handles zeros naturally; MA7 is robust to zeros without imputation.
- **Simplicity → adoption:** MA7 = “average last week × (LT+R)”. A bar manager can verify on a calculator; ML needs retraining, feature drift monitoring, and GPU-free but still opaque.
- **Empirical edge small:** ML-RF beats MA7 by only 5–8% (590 vs 620 MAE). The lift does not justify complexity for v1; we log it for A/B test in week 5.

Inventory model: Base-stock Par = $\mu(LT+R) + z\sigma\sqrt{LT+R}$ (base-stock / newsvendor). Chosen over EOQ because demand is variable and stockout cost >> ordering cost in hospitality; EOQ assumes deterministic demand. Min-max simulation validated.

4. How does the system perform? What would we improve?

Forecast accuracy (chain total, rolling-origin backtest):

- 7 d: **MA7 MAE 620 ml (≈ 0.8 bottle) MAPE 11%**; ML-RF 590 ml /10.2%
- 14 d: MA7 710 ml /12.5%; ML-RF 680 ml /11.5%
- Per-SKU median: MAE ~ 180 ml, MAPE $\sim 35\%$ (inflated by zero-days; aggregated planning matters).

Inventory simulation (30 days post 2023-10-14, actual demand as ground truth):

Metric	Status quo (order 1,239 ml when empty)	Proposed (Par/ROP)	Δ
Chain service level	76% (7/30 stockout days)	96% (1/30)	+26% availability
SKU-aggregate stockout-days (96 SKUs)	124 / 2,880 SKU-days (4.3%)	39 / 2,880 (1.4%)	↓68%
Lost demand	64 L	18 L	↓71% (≈ 45 L saved)
Avg inventory	18.2 L chain	20.4 L	+12% holding
Turnover	9.1 $\times$ / month	8.3 $\times$	slight dip, acceptable

Interpretation: For $\sim 12\%$ more inventory (mostly safety stock on high-CV SKUs like Heineken, Bacardi), we cut stockouts by two-thirds. At ₹300 margin/bottle vs ₹5 holding/bottle/day, break-even is 0.5% service gain — we deliver 20 $\times$ that.

Limitations & next improvements (ranked):

1. **Hierarchical reconciliation** (`scikit-hts`): Borrow strength across SKU $\rightarrow$ Type $\rightarrow$ Bar levels $\rightarrow$ expect $\sim 10\%$ MAPE on sparse SKUs.
2. **Censored demand correction:** Ingest POS "out-of-stock" button + EM to inflate demand on stockout days.
3. **Cost-based Z:** Replace fixed 95% with newsvendor critical ratio $C_u / (C_u + C_o)$ per SKU (margin vs spoilage cost).
4. **Probabilistic forecast:** Quantile regression (P95) outputs Par directly without normality.
5. **External regressors:** Occupancy, events, weather, IPL calendar, weekday (DOW already included).
6. **Stochastic lead time:** $SS = Z / (LT\sigma_D^2 + \mu^2\sigma_{LT}^2)$ for supplier variability.

What would break at scale & Production Tracking:

- **What breaks:** 28-day moving window fails if COVID/lockdown happens or major menu swap occurs (cold start problem for new SKUs). Also, `scipy` / `pandas` memory might blow up if scaling from 6 bars to 600 bars. We'd need to port the logic to PySpark or a dbt SQL pipeline.
- **What to track:** Track per-SKU MAPE drift $>15\%$ $\rightarrow$ auto-alert DS; track service level (in-stock %), spoilage waste %, and Days of Supply on Hand (DOSH) weekly.

5. How would this solution work in a real hotel?

08:00 daily cron (Airflow)

POS + Inventory DB → ETL (Python/pandas: clean, per-SKU daily agg) → Forecast (MA7 primary, ML-RF challenger)

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Supplier API ← P0 (qty = Par - Close) ← Par/ROP Engine (LT=3, Z=1.65, 28-d window)

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Dashboard (Streamlit/Grafana) +

Slack/Email alerts

Bar manager: table SKU | Par | ROP | Current | Order Qty | Reason → Approve → auto-P0

GM: weekly KPIs (service, stockouts, holding, forecast MAPE)

Tech footprint: Python + pandas + cron + Streamlit/Grafana, Snowflake/GSheets source, no GPU, <1 min retrain for full chain on laptop. Model registry logs MAE per SKU; challenger promoted if beats MA7 two weeks running.

Workflow example: 2023-10-14, Brown's Bar × Yellow Tail window $\mu=106$ ml/d, $\sigma=225$ ml → SS=742 ml, Par=1,250 ml, ROP=1,000 ml. Current close 0 → ORDER 1,250 ml. Order placed 14th, arrives 17th. System would have prevented the 2 stockouts that status-quo suffered that week.

Rollout: 4-week pilot on 1 bar (Anderson's) vs control bar; expand to 6 if stockout-days $\downarrow \geq 30\%$ (simulation predicts 68%). Feedback loop: after delivery, audit updates Opening; drift >100 ml triggers recount; "item unavailable" taps feed back to de-censor demand.

Business ROI (illustrative): Chain loses ~64 L/30 d = ~85 bottles → ₹25,500 margin lost. Proposed loses 18 L → ₹7,200. Holding extra 2.2 L costs ~₹330. **Net saving ≈ ₹18,000/30 d per chain**, plus NPS lift.